

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS CS candidate (NC State, May 2026) with hands-on production experience building full-stack TypeScript/React/Next.js applications and backend APIs. Shipped 10+ features across 7 production deployments at an AI-native startup -- including real-time event pipelines, authentication systems (Auth0/RBAC), PostgreSQL-backed REST APIs, and document processing workflows. Built real-time data platforms with Redis, SSE, and Next.js; fluent in AI coding tools (Claude, GPT-4o, Gemini, MCP) as part of daily development workflow.

CORE COMPETENCIES

TypeScript + React + Next.js

Tailwind CSS + Component Systems

REST API Design + PostgreSQL

Authentication (Auth0 / OAuth 2.0 / RBAC)

Real-Time Data (SSE / Redis / Event Pipelines)

Playwright + Jest + Testcontainers

AI Tooling (Claude, GPT-4o, Gemini, MCP)

Docker + GitHub Actions CI/CD

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern -- Raleigh, NC

- Developed **10+ backend features and RESTful APIs** in TypeScript, Node.js, Express, and PostgreSQL for an AI-powered claims automation platform -- 7 production deployments, zero regressions
- Implemented **OAuth 2.0 authentication via Auth0** with role-based access control, real-time event pipelines using Redis and Pusher, and document processing pipelines (PDF generation, email parsing via Microsoft Graph API, S3 file storage)
- Integrated multi-model **AI agent tooling** across 3 providers (Anthropic Claude, OpenAI GPT-4o, Google Gemini) using MCP, with Temporal.io workflow orchestration for agentic pipelines
- Shipped **Playwright automated testing**, Sentry observability, Docker containerization; contributed 394 commits and participated in full PR review cycle on a 2-person team

NC State University April 2025 - Present

Graduate Research Assistant -- Raleigh, NC

- Designed a cloud-based data platform serving 500GB genomic datasets to 100+ researchers on Google Cloud Run
- Collaborated with ML research team on model architecture design and time-series analysis pipelines; improved model accuracy by 12%

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern -- Remote

- Built 5+ AI-driven automation workflows including an OCR/NLP/ML health claim pipeline; improved accuracy by 25%
- Automated deployments on AWS and GCP; integrated Grafana monitoring dashboards; supported model validation

Capgemini Jun 2023 - Aug 2023

AI/ML Intern -- Remote

- Built distributed data pipelines using Hadoop + Spark and Kafka real-time streaming, reducing processing latency by 20%

PROJECTS

PulseRank -- Real-Time Leaderboard & Activity Feed

Next.js + Real-Time Data

Built a **real-time data platform in TypeScript + Next.js**: MySQL for transactional event writes with idempotency-key deduplication and outbox pattern, MongoDB for feed projections, **Redis sorted sets for sub-millisecond leaderboard reads**, and live score updates pushed over Server-Sent Events. Validated data consistency with Testcontainers; load-tested with k6; shipped OpenTelemetry + Prometheus observability and GitHub Actions CI/CD.

Stack: TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis, Tailwind CSS, Docker | github.com/Aditya-k24/PulseRank

AlgoPulse -- Agentic AI Platform

Full-Stack Product

Architected a full-stack AI orchestration platform with a **React + TypeScript** client and event-driven backend. Identified an API-polling bottleneck and redesigned around Supabase Edge Functions streaming AI reasoning loops directly to clients. Scaled worker architecture with Temporal.io and Kafka -- 2,000 async inference requests/second without state loss.

Stack: TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka | github.com/Aditya-k24/AlgoPulse

Inventory Reservation Service

Backend API + PostgreSQL

Built a transactional inventory reservation API in Java 21 + Spring Boot preventing overselling via **PostgreSQL row-level locking** (SELECT FOR UPDATE). Idempotent endpoints with RFC 9457 error responses, Flyway migrations, Spring Security RBAC, multi-stage Docker build, and GitHub Actions CI. Verified concurrency correctness with Testcontainers -- two concurrent reservations against quantity 1 produce exactly one success and one 409.

Stack: Java 21, Spring Boot 3.5, PostgreSQL, Flyway, Testcontainers, Docker | github.com/Aditya-k24/InventoryReservationService

LLM Eval Harness -- Vendor-Neutral Benchmark

AI Tooling

Built a benchmark comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro across 2,000 frozen test examples -- tracking ExactMatch, TokenF1, hallucination rate, p50/p95 latency, and cost/call. Statistical significance via paired bootstrap CIs and McNemar test. Demonstrates practical daily use of AI coding tools at depth.

Stack: Python, Anthropic/OpenAI/Google SDKs, DuckDB, Parquet, Streamlit, Docker | github.com/Aditya-k24/llm-eval-harness

EDUCATION

MS Computer Science -- North Carolina State University	Aug 2024 - May 2026
Raleigh, NC Courses: Artificial Intelligence, Operating Systems, Data Science, Algorithms, Software Engineering	
B.Tech Computer Science -- Mumbai University	Aug 2020 - May 2024
Mumbai, India CGPA: 9.32/10 Courses: Big Data, Machine Learning, DS & Algorithms, Software Engineering	

SKILLS

Frontend: TypeScript, React, Next.js, Tailwind CSS, NestJS, Storybook, Playwright, Jest, Vitest
Backend: Node.js, Python, Java, PostgreSQL, MySQL, MongoDB, Redis, Prisma, Express, REST
AI/ML: Claude API, OpenAI API, Gemini API, MCP, LangChain, RAG, TensorFlow, PyTorch, Hugging Face
DevOps: Docker, Kubernetes, GitHub Actions, AWS, GCP, Grafana, OpenTelemetry, Prometheus, Sentry
Auth: Auth0, OAuth 2.0, RBAC, JWT, Clerk